Course Project Final Project Report

CSE 578: Data Visualization

Casey Tom

Arizona State University

ctom4@asu.edu

Goals and Business Objective:

This report aims to present the findings and insights gained from the analysis to address questions posed by the UVW College's marking team. UVW College is looking to bolster its enrollment by targeting specific demographics based on income level. To support this effort, UVW College's marketing team has enlisted XYZ Corporation to assist them with this project. UVW aims to focus on individuals earning less than $50,000 annually as the primary criterion for marketing its degree programs. Our task at XYZ is to work with UVW using U.S. Census Bureau data to determine the key demographics to develop marketing profiles to target in their marketing campaign. The dataset includes key variables that help assess whether individuals earn less than or more than $50,000, such as age, work class, education level, marital status, and occupation.

To achieve its enrollment target, the marketing team at UVW would like XYZ to identify the factors determining whether an individual earns above or below $50,000. XYZ will develop the application based on research identifying the important factors influencing salary so they can use it to efficiently target the correct demographics for marketing their college to potential students.

Assumptions:

Business

- UVW College has chosen the $50K salary as the marker for lower and higher salary boundary for the marketing campaign.
- The UVW marketing department would like to create profiles based on factors such as age, gender, education ethnicity, income, etc.
- In order for UVW College enrollment expansion marketing to work, getting a college degree must have a beneficial to the students. This benefit is represented by increasing the chance to make over $50K salary.

Technical

- The data source for this analysis is a biased extract from the 1994 US Census database.
- When we initially explored the data, we noticed that some categories had more data points than others. The unevenness of the samples across categories makes using raw data problematic. The groups with higher representation will dominate the derived statistics. Due to the unevenness of the sample data across categories, most of the graphs use proportions or percentages within the groupings.
- Categories such as age, education and marital status will be recategorize to create less smaller categorial groups. This allows for graphs to be less crowded without losing important information.

Users Stories

1. Does receiving more education give a better chance of obtaining a higher income?

The UVW market team asked if more education gives a better chance of obtaining a higher income. This is the most important question we need to answer because it is the cornerstone of our marketing strategy. The marketing effort rest on the assumption that taking the time and effort to get more education will give a higher probability of earning more in the future. In this marketing campaign, we will use a salary of more than 50K to indicate higher earnings.

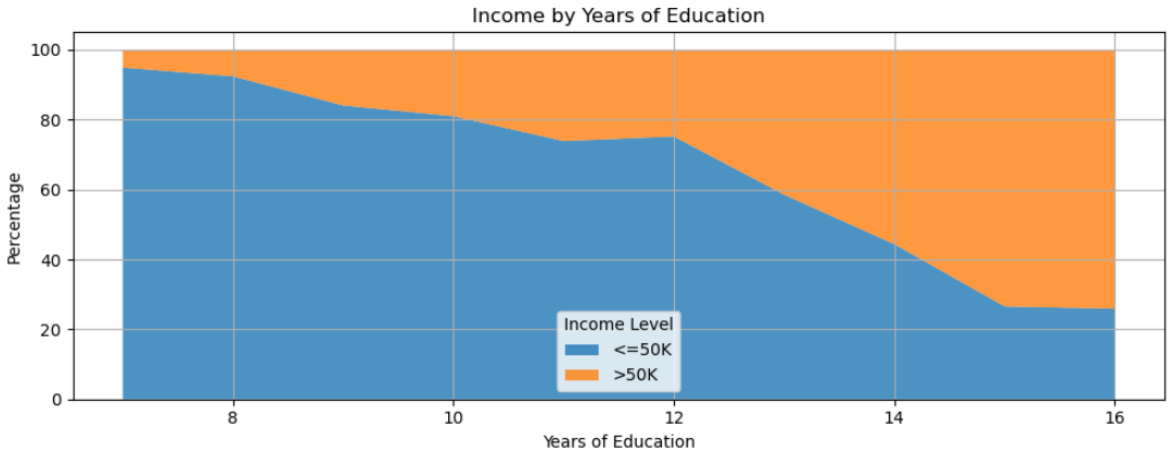


Figure 1

Figure1 shows there is a relationship between the number of years a person went to school and their income potential. From the area graph, we can see the percentage of people making over $50K, orange area, is close to zero when a person receives less than 8 years of education while a majority of those having 14 years or more years of education earn more than $50K a year. The area graph was chosen due to the ease of seeing the change in the percentage of people earning over and under $50K as the years of education increase.

2. Is there an income difference between the sexes for the same number of education years?

The UVW marketing team asked if there was an income difference between the sexes when controlling for the same amount of education years. As we can see, higher income is a benefit relative to years of education. We need to drill down to additional details of the demographics relating to years of education and income. The reason this is important is that to be more efficient in our demographic-specific targets, we need to find the demographic with the largest pool that can benefit from additional years of education.

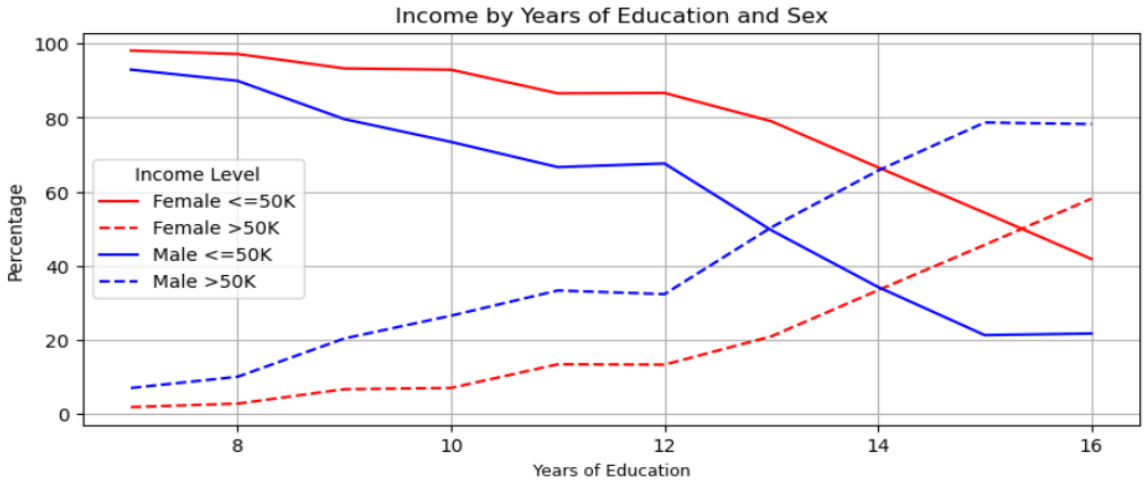


Figure 2

Figure 2 reinforces the relationship between years of education and income. The graph in Figure 2 gives us additional information broken down by sex. We can see females, regardless of their education, made less income than males when compared to the same years of education. The solid lines in the graph show the percentage making less than $50K; we can see males, blue lines, have a smaller percentage making less than

50K compared to females, red line. We can also see the same pattern in the dotted lines, representing income over $50K. The dotted line also shows that for the same years of education, a more significant percentage of males making more than $50K compared to females.

We should target both sexes between 11 and 12 years of education to attend college since a majority of that demographic makes less than $50K. The reason marketing to 11 to 12 years of education is they are either still in high school or have graduated and have not continued to college. This population has a high percentage of people who do not make $50K and are college-ready. Marketing should emphasize female candidates because this demographic represents a significant percentage not earning more than $50K.

3. Does income increase with age?

As we try to understand better the demographics that can benefit from continuing their education, UVW marketing asked if income increases with age. This question will also help us identify potential UVW students. The graph in Figure 3 answers the question about the relation between age and income. We can see the orange area increase in the mid-20s to their 50s and then decline. This increasing income represents the percentage of people earning over $50K. Health and retirement influence the likelihood of a drop off in the mid-50s and into their 60s. Health usually declines as people age, leading to the ability to work or work longer hours. Also, a factor might be that people start to retire in their mid-to-50s and accelerate into their 60's. The data we have cannot validate the cause of the income decline discussed, and we will have to find other data sources to investigate this theory.

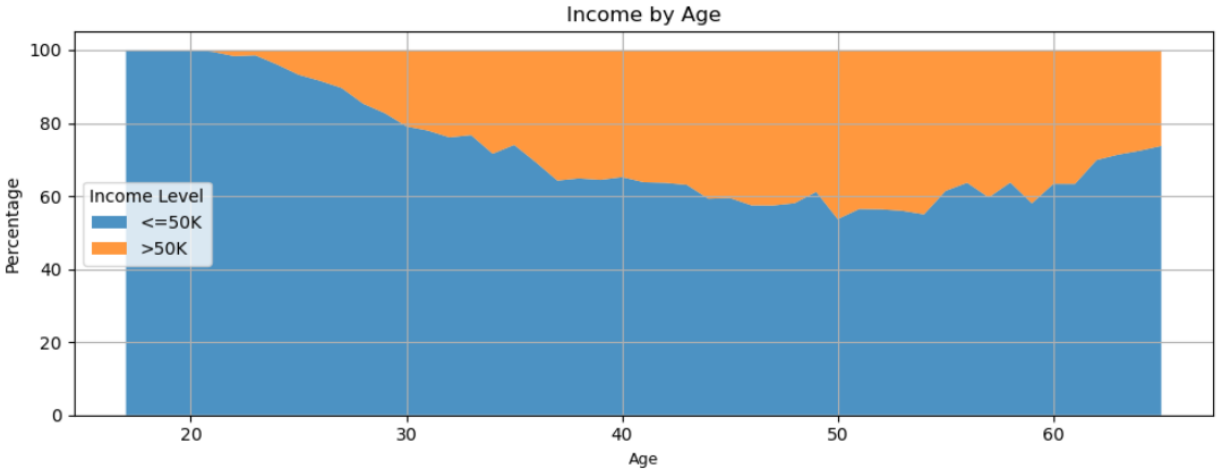


Figure 3

The reason we asked about the relationship of age to income is important because knowing younger people in the population generally earn less; the younger population should be a target for the UVW College marketing team. Now that we know younger people make less money, we need to see if we can get additional data on which demographic within the target within the younger age group.

4. What is the relationship between age, sex and number of hours work per week?

The UVM marketing team felt younger people might have more time to return to school. The marketing team asked if we could get information on the hours worked by age and sex. They feel that younger people below 30 or maybe 35 should have the bandwidth and wanted us to verify if this is true.

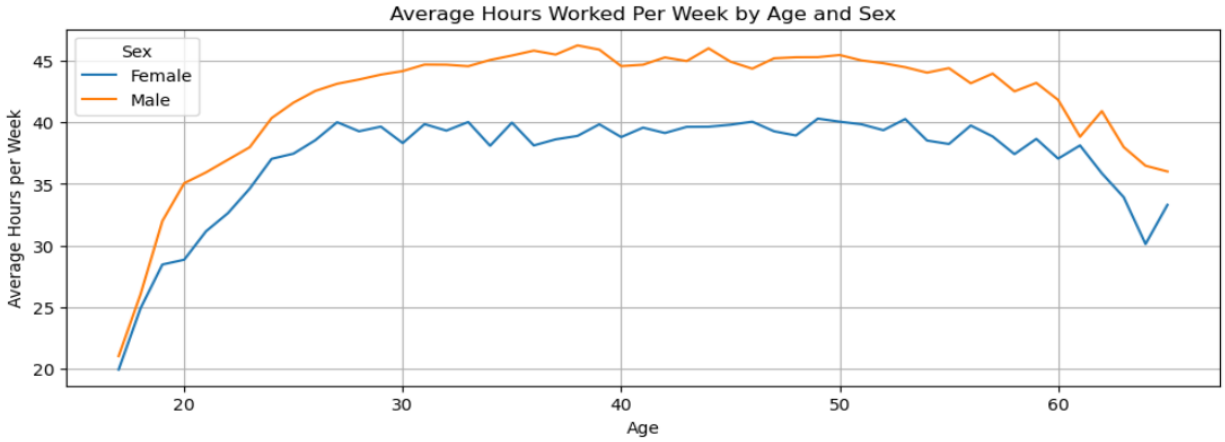


Figure 4

Figure 4 shows that regardless of sex, the average hours worked ramp up from their teens to around 30 years old, which is relatively stable until their mid-50s. On average, males work more hours per week than females. If the UVM wants its marketing team to target people who might work and go to school, it should look for people from their late teens to mid-20s. In this age group, females will have less of a workload to go to work and attend college.

5. Does being married or not married make a difference regarding income?

The UVM marketing team wanted to know if marital status influences income when normalized by the amount of education. To answer this question, we converted our education field to a smaller number of categories with the mapping shown in Table 2. Since our study aims to market to potential students, we removed the education category Lower Grade from this study. We also mapped the marital status field to either married or not married with the mapping shown in Table 3. The mapping is to reduce the number of categories but not lose the relevant information. This category reduction allows for clear summary information without losing the relevant data.

Eductation Map	
education	education_category
Assoc-acdm	Assoc
Assoc-voc	Assoc
Bachelors	College Grad
Doctorate	College Grad
Masters	College Grad
HS-grad	HS Grad
10th	Lower Grade
1st-4th	Lower Grade
5th-6th	Lower Grade
7th-8th	Lower Grade
9th	Lower Grade
Preschool	Lower Grade
Prof-school	Prof-school
Some-college	Some College
11th	Upper HS
12th	Upper HS

Table 2

Maritial Status Map	
marital_status	marriage_category
Never-married	Not Married
Married-civ-spouse	Married
Divorced	Not Married
Separated	Not Married
Married-AF-spouse	Married
Widowed	Not Married
Married-spouse-absent	Married

Table 3

College Status Map	
education	College Status
Bachelors	College Grad
HS-grad	Not College Grad
11th	Not College Grad
12th	Not College Grad
Masters	College Grad
Some-college	Not College Grad
Assoc-acdm	Not College Grad
Assoc-voc	Not College Grad
Doctorate	College Grad
Prof-school	College Grad
7th-8th	Lower Grade
9th	Lower Grade
5th-6th	Lower Grade
10th	Lower Grade
1st-4th	Lower Grade
Preschool	Lower Grade

Table 4

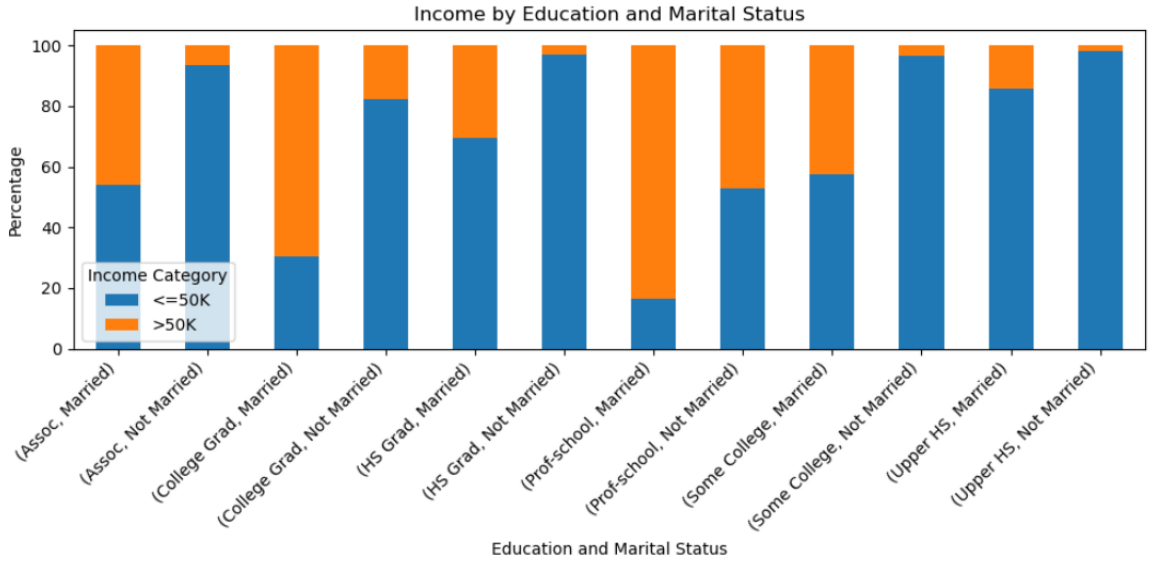


Figure 5

The result from using the reduce category and its impact on income can be seen in Figure 5. The bars for same education category are arranged next to each other with the first bar being Married and the next one Not Married. The graph shows for all education categories, the married group, orange portion, on average makes more than the not married group. This information is another datapoint for the marketing team to target their campaign toward individual that are not married.

6. By occupation, is there a income difference between college graduates and non-graduates?

The UVM marketing team asked by occupation if there is a difference between college graduates and non-graduates. To answer this question, we used the mapping in Table 4, as shown earlier, to map the values in the education column to the college status. The statuses we cared about were College Grad and Not College Grad. We filter out the Lower Grades as before since they will not be the demographic we will go after.

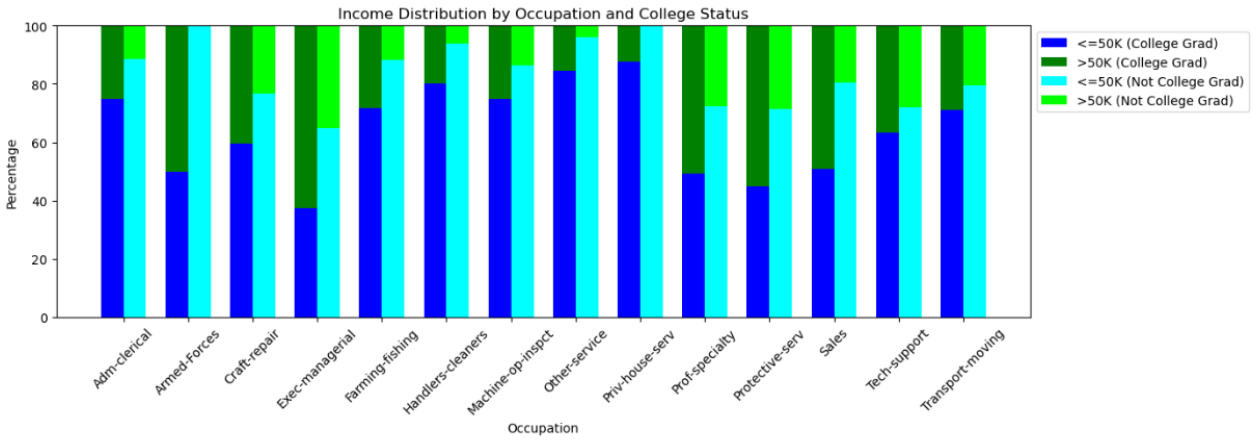


Figure 6

The graph in Figure 6 shows college graduates make up a higher percentage of greater than $50K versus non-college graduates in all occupations. This finding gives a more detailed look into our first question, where we investigated whether more education years relate to higher income. From this information, we should mention these statistics in the marketing material to emphasize the benefit of a college degree.

To create this chart in Figure 6, we needed to merge the $50K salary with the College Grad and Not College Grad data into a new category. Now we group the Occupation data with merged salary and college status to create the stack bars and to have bars next to each other. The format of Figure 6 is easier to see the grouping

relationship than Figure 5 due to explicit display of the category being group closely to each other. The comparison of the 2 ways to chart the data gives us a view on how a subtle difference can enhance the presentation.

Summary:

The cornerstone for the new marketing effort by UVW College will be based on the benefit of higher education and its correlation to higher income. The UVW College team felt that in order for the new marketing effort to succeed, students must have a positive benefit component to attend college and get a degree. Our first investigation was to establish if there was a correlation between years of education and the ability for a higher income. Figure 1 shows a clear relation between having more years of education and the potential for an income over $50K. This graph showed that the percentage of people with an income of more than $50K was nearly zero compared to the majority with 14 or more years of school. With the confirmation between education and income, the following steps are to understand the demographics in detail in order to better create a profile for the new marketing effort. Figure 2 highlights the income discrepancy between the sexes. It showed males across all education years with a higher percentage of earners over $50K annually. Figure 3 shows that younger people earn less than older ones. The graph shows that the percentage of people under 35 who make over $50,000 annually is much smaller than those over 35, making the under 35 a key demographic to target for the new marketing campaign. Figure 4, showing the relationship between age, sex, and the average hours worked per week, reinforces that the younger population should be the emphasis of the marketing target due to lower hours worked. Figure 4 also reinforces that females tend to work fewer hours than males across all ages. Figure 5 indicates that married people earn more than unmarried. This pattern of married earning more holds for all education levels, giving us another factor for UVW College to target in their campaign. Finally, Figure 6 shows college graduates earn more than non-graduates in all occupations and should be highlighted in the marketing material as one of the reasons for attending UVW.

There are other factors we have not investigated, such as native country, race, and personal relationships as it relates to income and education. Other aspects we can dive into in the future are the distribution of age and sex as factors in occupation versus income.

Recommendations:

Based on the research presented, the marketing material should communicate the benefits of a higher education as it relates to income. In addition, marketing should emphasize that in all occupations, people with a college degree are a higher percentage of those earning more than $50,000 annually.

The marketing demographics should target younger, late teens to 30 years old, unmarried individuals without a college education, with emphasis on females. This group has a lighter workload and a larger proportion of individuals making less than $50K. This demographic can most benefit from a college degree and with the bandwidth to attend school.